VoiceOps 2026

Final Integration & Submission Candidate Report
Status: INCOMPLETE_INTEGRATION_TESTS
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Project overview

VoiceOps is a voice-capable customer-support agent for ZentroMart. It combines section-based knowledge retrieval, deterministic business rules and tool workflows, language-aware response handling, and optional LLM and speech-provider integrations. Retrieval ranks policy/help/FAQ sections and separates catalog results; application logic enforces verification, escalation and policy constraints.

Candidate and retained changes

This package reflects the v9.1.2 candidate after the two controlled rollback experiments. The HI_MAP extras and context_query call-site change were selectively reverted; other v9.1.2 work remains, including optional hybrid-retrieval plumbing, generic prompt updates, preflight model filtering, and request-budget enforcement. The two retrieval rollbacks did not change measured recall. No new runtime patch was introduced for this release.

Offline validation

Evaluation	Count	Content / checks	Decision	Recall	Outcome
Normal questions	52	100.0%	100.0%	0.968	PASS
ASR questions	52	100.0%	100.0%	0.968	PASS
Conversations	96 turns	100.0% checks	100.0%	0.944	23/23 fully correct

Compilation: exit 0. Full pytest: 62 passed. Dataset validator: exit 0 (9 CSVs, 21 KB documents, dataset integrity checks passed). Offline accuracy gate: PASS.

Retrieval baseline limitation

Compared with verified saved offline baselines, recall remains lower: normal 0.989 → 0.968; ASR 0.989 → 0.968; conversations 1.000 → 0.944. Content and decision accuracy remain 100% in the offline question evaluations; all 23/23 conversations are fully correct. Neither HI_MAP rollback nor the context-query rollback recovered the recall decline. The held-out diagnosis was highly templated and is not considered robust evidence for retrieval quality.

LLM/RAG evaluation

Run	Count	Content	Decision	Recall	Status
Normal RAG	52	98.1%	96.2%	0.968	Exit 0
ASR RAG	52	100.0%	94.2%	0.968	Exit 0
Conversations RAG	96 turns	100.0%	100.0%	0.944	23/23

The earlier ASR RAG attempt ended with a provider transport error and explicit offline fallback; it is invalid RAG evidence. The latest bounded ASR RAG run exited 0 but decision accuracy was 94.2%, below the prior reported RAG value; this is reported as observed, not treated as an offline result. RAG request ledger: 168/400 before these tests, 174/400 after; no ledger reset.

Gnani speech smoke test

Credential configuration was confirmed using boolean-only checks. TTS produced a non-empty 399,404-byte mono WAV at 48 kHz on the final allowed TTS attempt. One preceding TTS invocation was interrupted by PowerShell stderr handling after

the provider logged synthesis start, so total TTS attempts are counted as two. STT was attempted once on generated synthetic speech and failed with a sanitized RuntimeError; no transcript text is included. Result: TTS succeeded on retry; STT failed; speech integration testing is incomplete.

Other integration checks

The complete 62-test suite passed, covering the repository’s local and mocked behavior. No live ticket/order actions, phone calls, or business-changing operations were performed. Embedding retrieval was not tested. No new RAG retries were performed; LLM automatic retries were disabled for these runs.

Reproducibility and release contents

Python: use the project virtual environment. Install listed dependencies from requirements.txt. Configure credentials locally in a developer-owned .env file (not distributed), then run the offline suite with VOICEOPS_NO_ENV_LOAD=1, DATASET_DIR pointing to the dataset root, and python -m pytest -q. Run the dataset validator with python alidate_dataset.py. Live provider tests are not required for offline validation.

Repository: https://github.com/chukka-venugopalam/voiceops

The release submission ZIP uses an explicit allowlist and excludes .env, .env.example, credentials, .venv, .git, dataset contents, hidden cases, held-out labels, evaluation JSONs, caches, request-budget ledgers, logs, backups and temporary audio. The approved demo is previously recorded synthetic media and does not establish that the current Gnani STT test passed.

This is a submission candidate, not a guaranteed winning score. Integration status is INCOMPLETE_INTEGRATION_TESTS because STT did not pass and the earlier ASR RAG run was invalid.